

Clinical: Electrolyte Disorders – Calcium, Magnesium and Phosphate

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Session Objectives

- Outline calcium regulation and dysregulation.
- Outline phosphate regulation and dysregulation.
- Outline magnesium regulation and dysregulation.

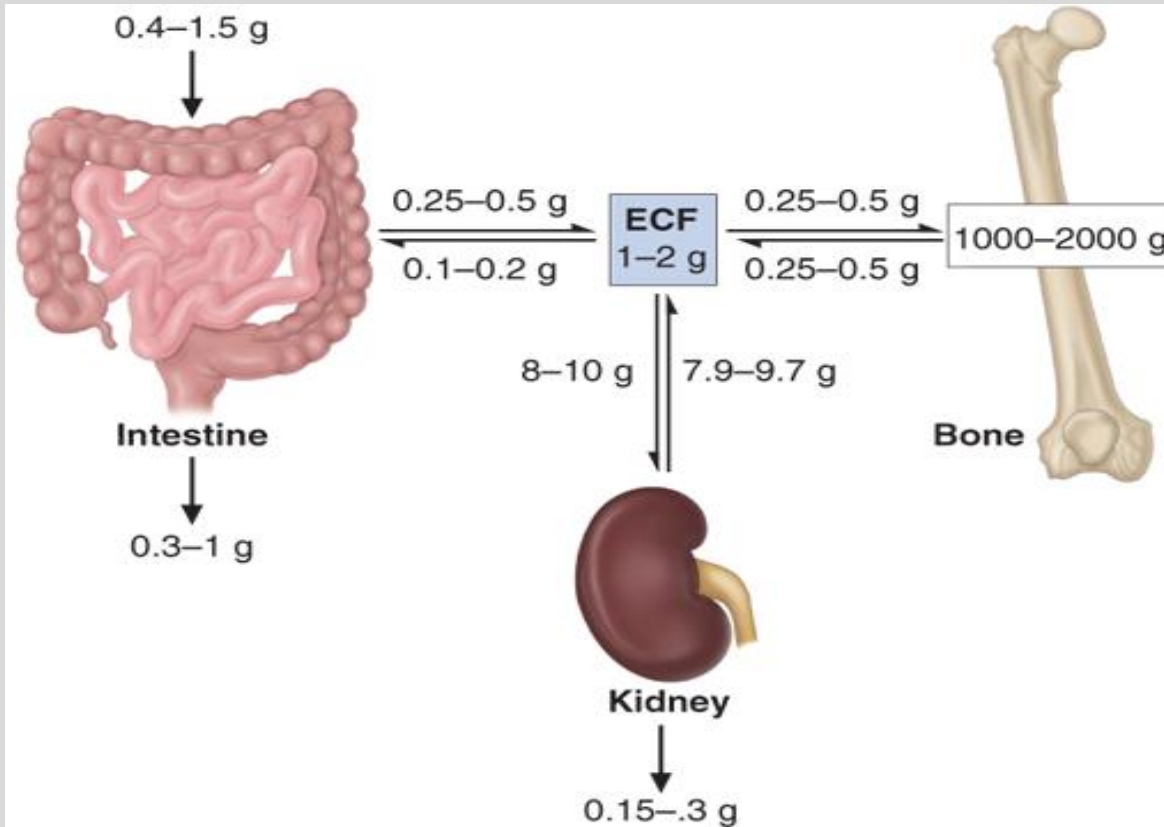
Calcium

- 40% bound to albumin
- 15% complexed with citrate, sulfate, phosphate
- 45% ionized (free)

Calcium Homeostasis

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Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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Calcium Handling

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- **Gastrointestinal**
 - Active – 20% to 70% -Transcellular TRPV6 in duodenum and jejunum
 - Passive -5% of intake - Paracellular throughout intestine
- **Bone**
 - 99% of body calcium
 - Osteoblasts vs. Osteoclasts

Calcium Handling

- Renal
 - Only ionized (non-protein bound) filtered
 - Reabsorption – 99%
 - 70% Proximal tubule
 - 20% TALH
 - 15% Distal Nephron

Calcium Regulation

- Parathyroid hormone
 - CaSR
- Increases plasma calcium
 - Intestinal absorption
 - Bone resorption
 - Renal reabsorption

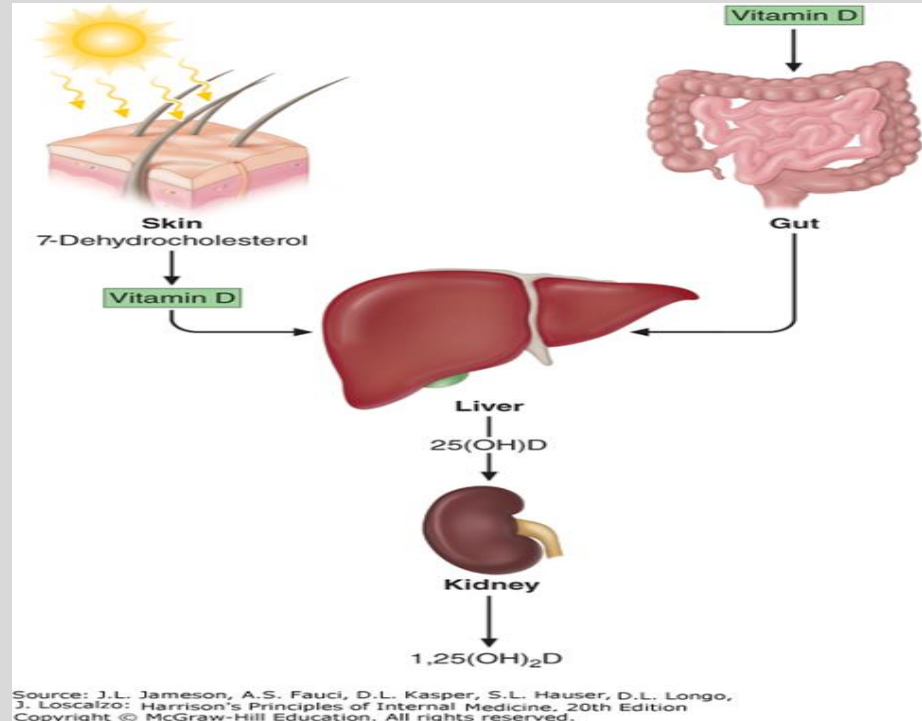
Vitamin D and Calcium

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Vitamin D

- Intestines
- Bone
- Renal

Calcium self regulation



Hypercalcemia Etiologies

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- Excess PTH production
 - Primary Hyperparathyroidism –
 - Tertiary Hyperparathyroidism
 - Ectopic PTH secretion
- Malignancy
 - PTHrP – Squamous cell carcinomas
 - Lytic metastases

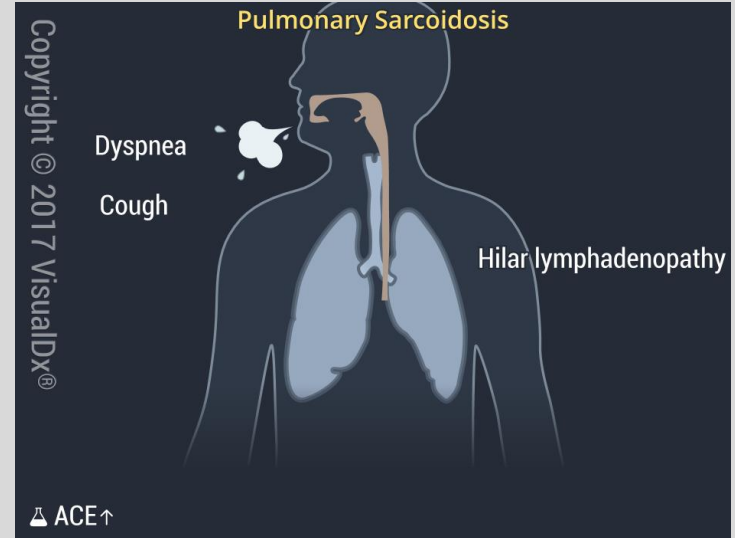


Hypercalcemia etiologies

- Hypervitaminosis D
 - Sarcoidosis
- Excess Ca intake
 - Milk alkali syndrome
- Increase in bone reabsorption
 - Immobilization
 - Hyperthyroidism

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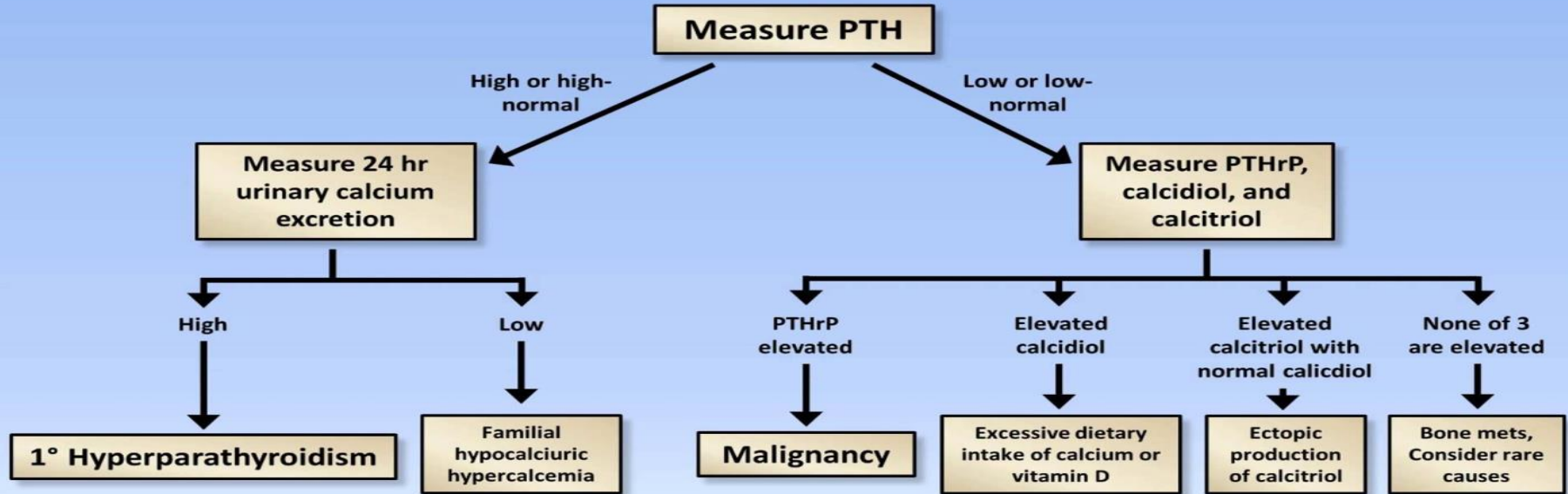
Clinical Manifestations

- Correct for albumin
- Mild 11–11.5 mg/dL - asymptomatic
- Moderate 12-14 mg/dL – Fatigue, muscle weakness/pain
- Severe >14 mg/dL
 - Neuropsychiatric
 - Gastrointestinal
 - Renal & Cardiovascular

Hypercalcemia Diagnosis

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Diagnostic Algorithm



Source: (2013). Hypercalcemia. *YouTube*.

Hypercalcemia Treatment

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- Mild to Moderate – avoid factors aggravating hypercalcemia.
Adequate hydration.
- Severe –
 - 4-6 L NS in first 24 hours
 - Calcitonin
 - Bisphosphonates
 - Glucocorticoids

Hypercalcemia Treatment

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- Treat underlying causes
- Hyperparathyroidism – parathyroidectomy if symptomatic. Asymptomatic based on risk of end-organ disease.

Hypocalcemia Etiologies

- Low PTH (Hypoparathyroidism)
 - Destruction of gland
 - Abnormal gland development
 - Altered regulation of PTH

Hypocalcemia Etiology

- High PTH (secondary hyperparathyroidism)
 - Vitamin D deficiency/resistance
 - Parathyroid hormone resistance syndromes
 - Extravascular deposition
 - Severe tissue injury

Clinical Manifestations

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- Tetany
 - Trousseau's and Chvostek's signs
 - Video:

<https://www.nejm.org/doi/full/10.1056/NEJMicm1110569>

- Seizures
- Cardiovascular
- Papilledema

Laboratory evaluation hypocalcemia

	PTH	Corrected serum calcium	Phos	Mag	25OHD	1,25 (OH) 2D	Creatinine
Hypoparathyroidism	Low	Low	Elevated	Normal	Normal	Normal or low	Normal
Activating mutation calcium sensing receptor	Normal or low	Low	Elevated	Normal	Normal	Normal	Normal
Hypomagnesemia	Normal or low	Low	Normal	Low	Normal	Normal	Normal
PTH resistance (pseudohypoparathyroidism)	Elevated	Low	Elevated	Normal	Normal	Normal	Normal
Vitamin D deficiency	Elevated	Low or normal	Low or normal	Normal	Low	Normal or high	Normal
Chronic kidney disease	Elevated	Low	Elevated	Elevated or normal	Normal or low*	Low	Elevated

* In individuals with concurrent nutritional deficiency.

Treatment

- Mild (7.5 to 8.0 mg/dL)/chronic hypocalcemia – oral calcium supplementation with vitamin D3
- Severe (≤ 7.5 mg/dL)/symptomatic – IV calcium 90 to 180mg then switch to oral
- Correct concurrent hypomagnesemia

Phosphate Handling

- 85% in bone mineral
- Intestines
- Bone -fibroblast growth factor 23 (FGF-23)
- Renal – proximal tubule

Phosphate Regulation

- Determined by rate of renal tubular absorption.
- Intestinal absorption highly efficient therefore regulated by urinary excretion.
- The fractional excretion of phosphate (ratio of phosphate to creatinine clearance) is generally in the range of 10–15%

Phosphate Regulation

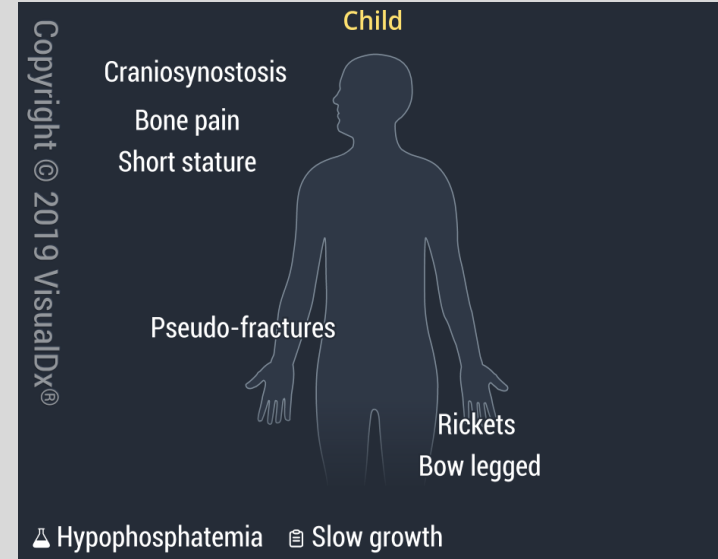
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- Reabsorption depends on dietary intake, serum concentration, PTH, Vitamin D3 levels and FGF-23
- PTH and FGF-23 reduce phosphate absorption
- Low intake and serum levels increase absorption
- Main factor stimulating PTH is calcium not phosphate

Reduced renal phosphate absorption

- Hyperparathyroidism
- Vitamin D deficiency
- Renal phosphate wasting – associated with rickets and osteomalacia
- Fanconi syndrome



Decreased intestinal absorption

- Inadequate intake
- Antacids with aluminum or magnesium
- Steatorrhea or chronic diarrhea

Shifts of phosphate into cells

- Increased insulin secretion
- Respiratory alkalosis

Clinical Manifestations

- Central nervous system
- Cardiopulmonary
- Skeletal and smooth muscle
- Hematologic

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Rhabdomyolysis. 2018. Elsvir Point of Care.

Diagnosis and Treatment

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24 hour urine phosphate excretion

- $<100\text{mg}$ – internal redistribution or decreased intestinal absorption
- $>100\text{mg}$ – reduced renal phosphate absorption

Treatment –

- Asymptomatic $<2.0\text{mg/dl}$ – oral therapy
- Symptomatic $1.0\text{-}2.0\text{ mg/dl}$ - oral therapy
- Symptomatic $<1.0\text{mg/dl}$ – IV therapy

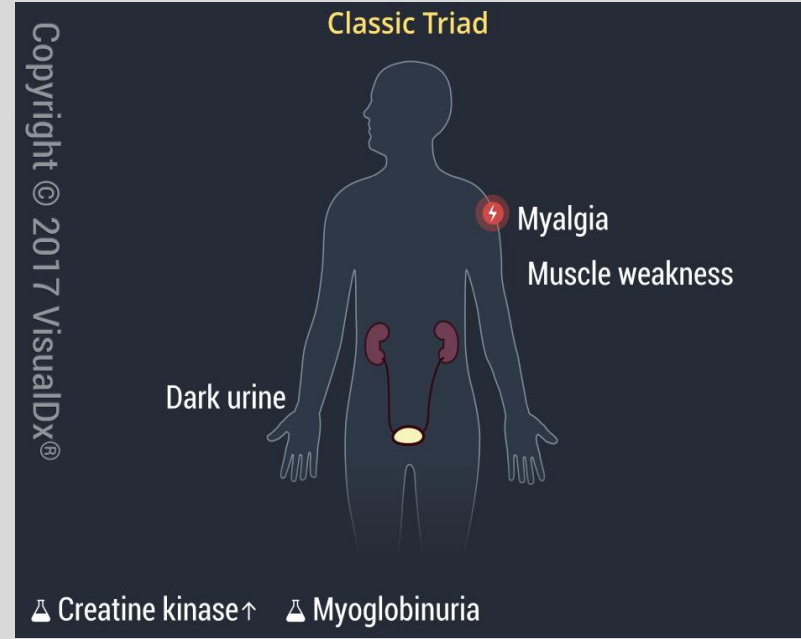
Impaired renal phosphate excretion

- Hypoparathyroidism
- Parathyroid suppression
- Renal insufficiency
- Acromegaly
- Familial tumoral calcinosis



Acute phosphate load

- Tumor lysis syndrome
- Rhabdomyolysis
- Exogenous phosphate

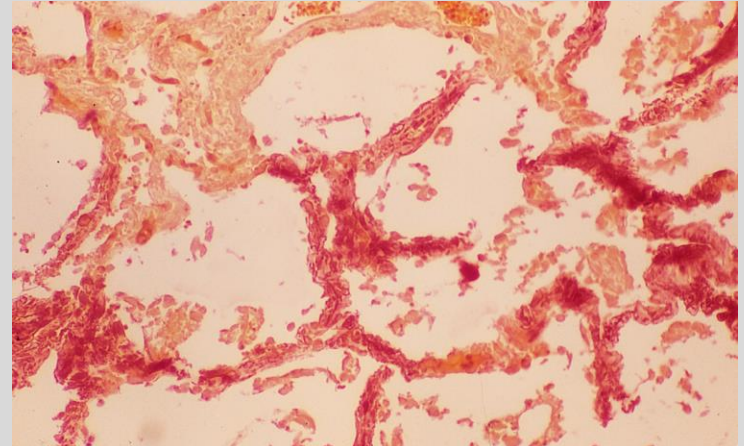


Clinical Manifestations

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- Tetany
- Seizures
- Renal failure
- Pulmonary/cardiac calcifications



Source: Michael A. Grieco, Jack A. Elias, Jay A. Fishman, Robert M. Kitzoff, Allan I. Pack, Robert M. Sano, Mark D. Siegel, Fishman's Pulmonary Diseases and Disorders, www.accessmedicine.com Copyright © McGraw-Hill Education. All rights reserved.

Treatment

- Limited therapeutic options
- Volume expansion
- Aluminum antacids
- Hemodialysis

Question

An 88-year-old male presents to ED unconscious. While taking his blood pressure you notice carpal spasm. Which of the following is the most likely diagnosis?

- A. Hyperkalemia
- B. Hyponatremia
- C. Hypercalcemia
- D. Hypocalcemia
- E. Lambert–Eaton myasthenic syndrome

Question

Lab studies are:

- Calcium 1.0 mg/dL (normal, 8.8-10.2 mg/dL);
- Phosphate 13.0 mg/dL (normal, 2.5-4.5 mg/dL)
- Parathyroid hormone 94.3 ng/L (normal 10-65 ng/L)
- 25-Hydroxyvitamin D 71 ng/mL (normal, 20-100 ng/mL)
- BUN 10 mg/dL (normal, 7-20 mg/dL)
- Creatinine 1.0 mg/dL (normal, 0.6-1.2 mg/dL)

What is the most likely diagnosis?

- A. Primary hyperparathyroidism
- B. Vitamin D deficiency
- C. Chronic kidney disease
- D. Squamous cell carcinoma of the lung
- E. PTH resistance syndrome

Question

A 55-year-old female presents to the ED with anxiety, constipation and abdominal pain. On CT of the abdomen, she is found to have nephrolithiasis. Which of the following is the most likely metabolic abnormality?

- A. Hypocalcemia
- B. Hyponatremia
- C. Hypophosphatemia
- D. Hypokalemia
- E. Hypernatremia

Labs are:

- Calcium 14.3 mg/dL (normal, 8.8-10.2 mg/dL)
- Phosphate 1.0 mg/dL (normal, 2.5-4.5 mg/dL)
- Parathyroid hormone 4.6 ng/L (normal, 10-65 ng/L)
- Parathyroid hormone-related peptide 3.8 pmol/L (normal < 2.0 p/mol)
- 1, 25-dihydroxyvitamin D 60 ng/mL (normal, 20-100 ng/mL)

What is the most likely diagnosis?

- A. Primary hyperparathyroidism
- B. Familial hypercalciuric hypercalcemia
- C. Multiple myeloma
- D. Squamous cell carcinoma
- E. Excess dietary calcium intake

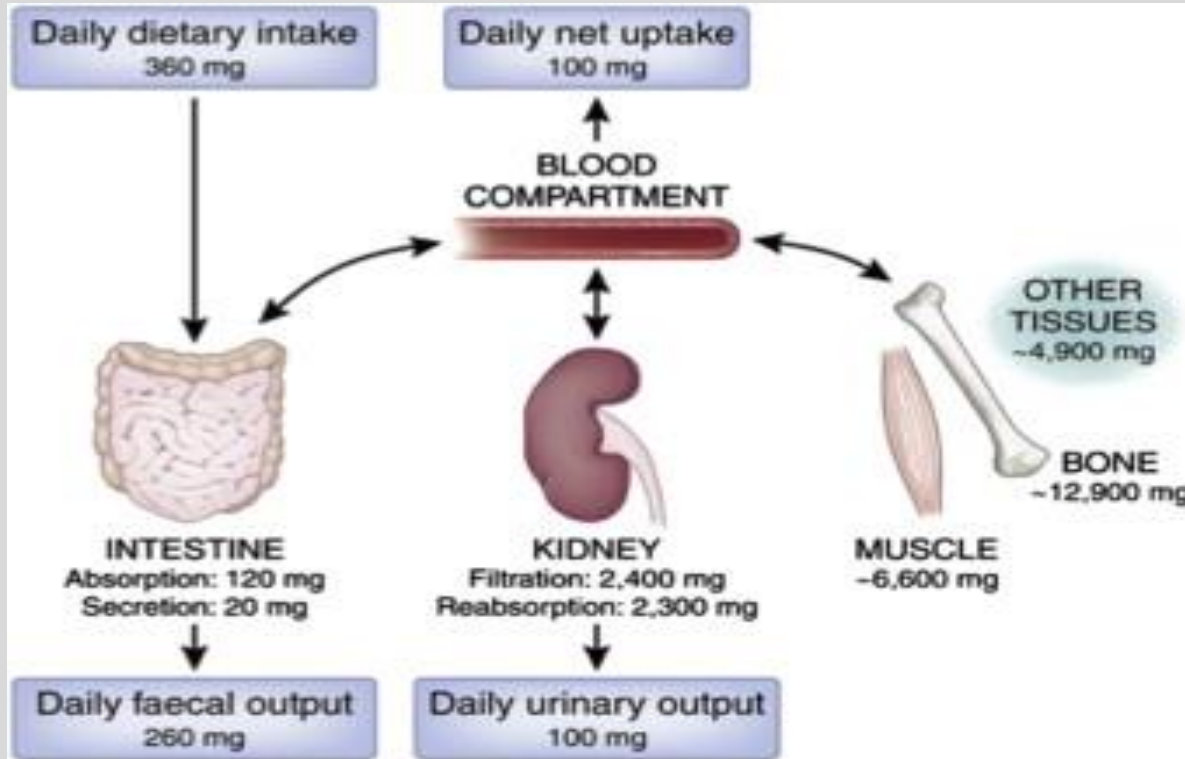
Magnesium

- Crucial for neuromuscular activity
- Forms key complex with ATP
- Concentration in serum closely regulated within the range of 0.7–1 mmol
- Only 1% of total magnesium in ECF

Magnesium Balance

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Source: Alhosaini M, Leehey DJ. Magnesium and Dialysis: The Neglected Cation. Am J Kidney Dis. 2015;66(3):523-31. DOI: 10.1053/j.ajkd.2015.01.029

Magnesium Handling

- Intestinal absorption
- Renal regulation
 - 70% thick ascending loop of Henle
 - 20% in proximal tubule
 - 10% in distal tubule

Magnesium Regulation

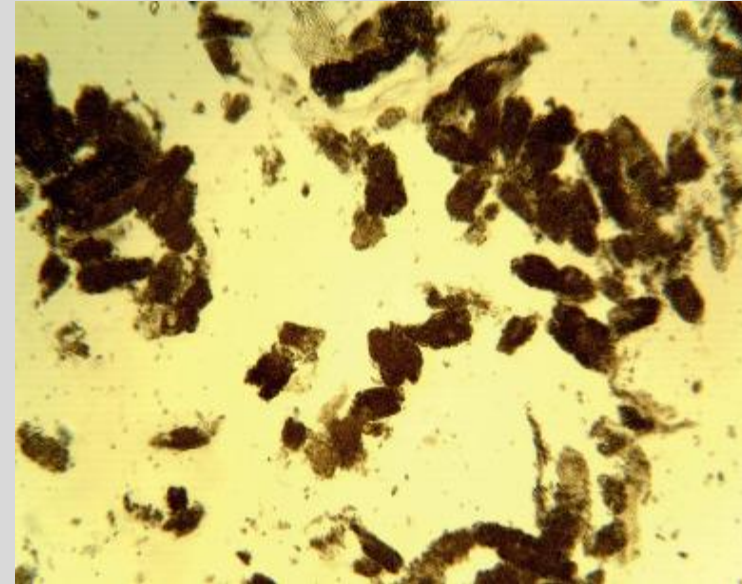
- Thick ascending limb
 - Passive
 - Junctional proteins
 - Increased reabsorption with PTH and hypomagnesemia
 - Decreased reabsorption with hypercalcemia or hypermagnesemia
- Distal convoluted tubule
 - Important for fine tuning

Intestinal Causes

- Diarrhea>vomiting
- Proton pump inhibitors
- Primary intestinal hypomagnesemia
- Hypomagnesemia with secondary hypocalcemia

Renal losses

- Genetic wasting syndromes
- Medications
- Tubular dysfunctions
- Uncontrolled diabetes
- Hypercalcemia



Hypomagnesemia Manifestations

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Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
www.accessemergencymedicine.com
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Diagnosis and Treatment

- If no apparent etiology check magnesium excretion.
- Mild asymptomatic – oral magnesium salts
- Severe – IV magnesium (i.e. torsades)
- Correct concurrent deficiencies

Hypomagnesemia			Normal	Hypermagnesemia			
Severe ← < 0.5	Moderate 0.5 - 0.9	Mild 1.0 - 1.4	1.5 - 2.0 mEq/L	Mild 2.1 - 5.0	Moderate 5.1 - 10.0	Severe 10.0 - 15.0	→ > 15.0 mEq/L
Cardiac arrest, seizure	ECG changes, tremor, spasm, cramping	Asymptomatic		Asymptomatic	ECG changes, dizziness, nausea, weakness	Paralysis, areflexia, low BP	Respiratory and cardiac arrest

Question

Which of the following are expected manifestations of hypermagnesemia?

- A. Decreased deep tendon reflexes
- B. Refractory hypotension
- C. Respiratory failure
- D. Facial flushing
- E. All of the above

Question

Which of the following cardiac complications are most likely to be present in hypermagnesemia?

- A. Tachycardia
- B. Short QT interval
- C. Short PR interval
- D. Heart block

Summary Slide

- Outline calcium regulation and dysregulation.
- Outline phosphate regulation and dysregulation.
- Outline magnesium regulation and dysregulation.
- Core References:
 - Bringhurst F, Kronenberg HM, Liu ES, Wein MN. Bone and Mineral Metabolism in Health and Disease. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. Harrison's Principles of Internal Medicine, 22nd Edition. McGraw Hill; 2026
 - Towler DA. Hypercalcemia and Hypocalcemia. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. Harrison's Principles of Internal Medicine, 22nd Edition. McGraw Hill; 2026
 - ScholarRx

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